

PES UNIVERSITY
Department of Computer Science and Engineering

Software Engineering — Lab 1
Problem Statement #06
Alumni Mentorship & Mock Interview Platform

Name: K R Manoj
SRN: PES1UG24AM125

1. Requirements Table

1.1 Functional Requirements (FR-001 – FR-005)

FR ID	Description	Priority	Acceptance Criteria (Pass / Fail)	Rationale
FR-001	The system shall recommend alumni mentors to students based on domain matching (e.g., Embedded Systems, Cloud Computing) and facilitate one-click session booking.	High	Pass: Mentorship session is added to both calendars with a meeting link. Fail: Double-booking of mentor slots occurs.	Reduces manual search effort for students and ensures matches align with relevant domain expertise, improving mentorship quality and engagement.
FR-002	The system shall atomically verify and lock a mentor's selected time slot at the moment of booking confirmation, rejecting any concurrent booking request for the same slot.	High	Pass: When two students attempt to book the same slot simultaneously, only the first request succeeds; the second receives a “slot no longer available” message with alternative slot suggestions. Fail: Both requests succeed, creating a scheduling conflict (double-booking).	Directly addresses the concurrency risk inherent in a shared mentor calendar; prevents scheduling conflicts that erode mentor trust and platform credibility.
FR-003	The system shall allow alumni mentors to record a structured post-interview scorecard (technical rating, communication rating, and free-text feedback) immediately after a mock interview session concludes.	Medium	Pass: Scorecard is saved, linked to the correct session, and visible to the student within 5 minutes of submission. Fail: Scorecard data is lost, mismatched to the wrong session, or not visible to the student.	Structured, timely feedback is the core value delivered to students; accurate linkage ensures trustworthy and actionable mentorship outcomes.
FR-004	The system shall allow alumni mentors to define, publish, and update recurring or one-off available time slots, which are reflected in real time to students during booking.	Medium	Pass: A slot added or removed by the mentor is immediately reflected as available/unavailable in the student's booking view. Fail: Stale availability data allows booking of a blocked, removed, or already-passed slot.	Accurate, real-time availability is the foundation that mentor matching, booking, and double-booking prevention (FR-002) all depend on.
FR-005	The system shall send automated confirmation notifications at booking time and reminder notifications to both student and mentor prior to the scheduled session.	Low	Pass: Both parties receive a confirmation immediately after booking and a reminder at least 30 minutes before the session. Fail: A notification is missing, significantly delayed, or sent to the wrong party.	Reduces no-shows and reinforces the reliability and professionalism of the overall mentorship experience.

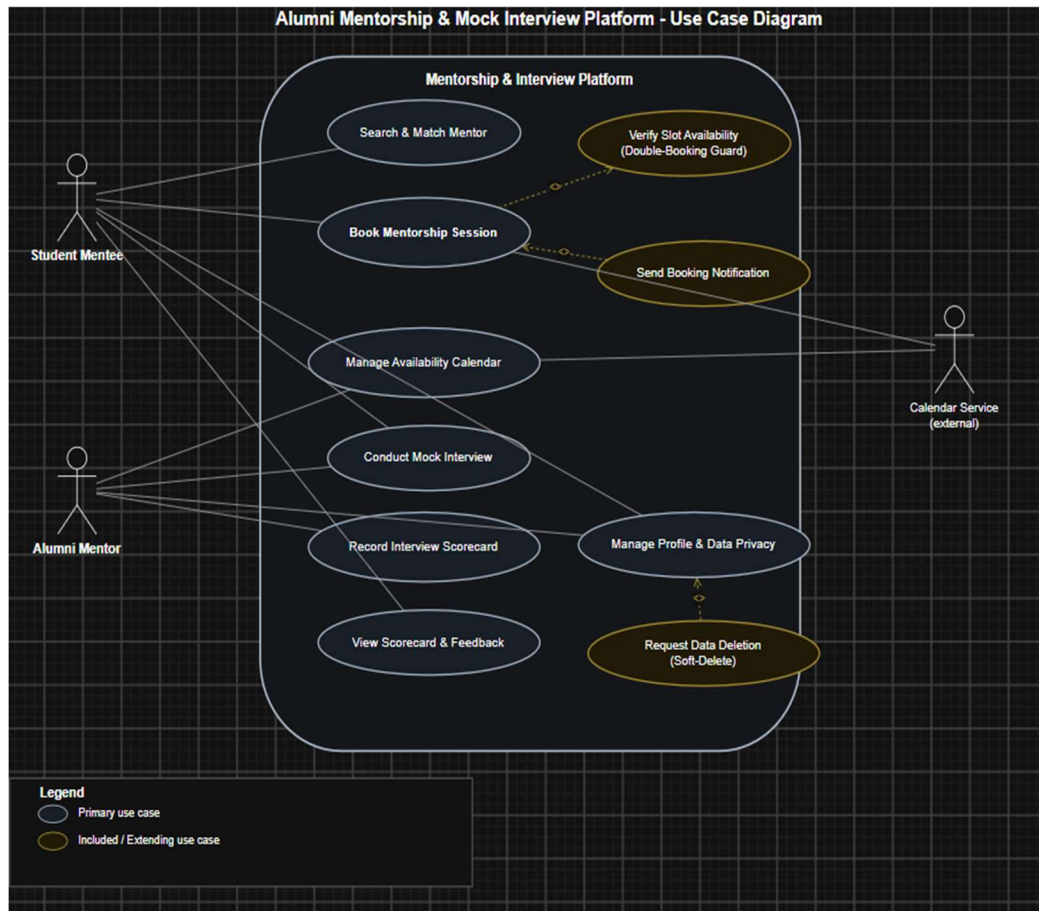
1.2 Non-Functional Requirements (NFR-001 – NFR-002)

NFR ID	Type	Description	Acceptance Criteria (Pass / Fail)	Rationale
--------	------	-------------	-----------------------------------	-----------

NFR-001	Performance & Security	All user profile data and feedback scorecards shall adhere to privacy guidelines and support soft-deletion within 24 hours of a deletion request.	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load, and deletion requests are honored (data marked inaccessible) within 24 hours. Fail: Latency/security benchmarks are missed, or deleted data remains accessible past 24 hours.	Compliance with data-protection expectations builds user trust and mitigates legal and reputational risk from mishandled personal data.
NFR-002	Reliability & Scalability	The booking subsystem shall use atomic, transaction-safe slot-locking so that concurrent booking requests (up to 500 simultaneous requests) are serialized without data corruption, while maintaining 99.9% monthly uptime for	Pass: Load testing with 500 concurrent booking attempts on the same slot results in exactly one successful booking and zero data corruption or deadlocks. Fail: More than one	Double-booking prevention (FR-002) is a correctness-critical, concurrency-sensitive operation; explicit, testable reliability guarantees ensure the business rule holds under

2. ML Use-Case Diagram

The diagram below models both actors, all primary use cases of the platform, and the required relationships: an «include» dependency (Book Mentorship Session includes Verify Slot Availability — the mechanism behind double-booking prevention) and «extend» dependencies (Send Booking Notification extends Book Mentorship Session; Request Data Deletion extends Manage Profile & Data Privacy).



Use-Case Flow Specification

UC-03: Record Post-Interview Scorecard

Use Case Information

Field	Details
Use Case ID	UC-03
Use Case Name	Record Post-Interview Scorecard
Primary Actor	Alumni Mentor
Supporting Actor	Student Mentee
Related Requirement	FR-005
Priority	Medium

1. Preconditions

1. The Alumni Mentor is registered and logged into the system.
2. A mock interview between the Alumni Mentor and Student Mentee has been completed.
3. The completed mock interview session is available in the system.
4. The Alumni Mentor has access to the post-interview scorecard functionality.

2. Postconditions

1. The interview scorecard is successfully recorded and stored.
2. Performance ratings entered by the Alumni Mentor are associated with the correct Student Mentee.
3. The mentor's qualitative feedback is saved along with the scorecard.
4. The recorded scorecard is available for the Student Mentee to view.

3. Main Success Scenario

1. The Alumni Mentor logs into the platform.
2. The system authenticates the mentor and displays the mentor dashboard.
3. The mentor opens the completed mock interview session.
4. The system displays the interview details.
5. The mentor selects the **Post-Interview Scorecard** option.
6. The system displays the structured scorecard form.
7. The mentor enters performance ratings for the Student Mentee.
8. The mentor enters qualitative feedback regarding the student's interview performance.
9. The mentor reviews the entered ratings and feedback.
10. The mentor submits the completed scorecard.
11. The system validates the scorecard information.
12. The system saves the scorecard successfully.
13. The system associates the scorecard with the completed mock interview and Student Mentee.
14. The Student Mentee can subsequently view the recorded scorecard and feedback.

4. Alternate Flow — Incomplete Scorecard

Condition: The mentor attempts to submit the scorecard without providing required information.

1. At Step 11 of the Main Success Scenario, the system detects missing required information.
2. The system displays a validation message indicating the incomplete fields.

3. The scorecard is not submitted or saved as a completed scorecard.
4. The mentor provides the missing information.
5. The mentor submits the scorecard again.
6. The system validates the information and continues with Step 12 of the Main Success Scenario.

5. Successful Outcome

The Alumni Mentor successfully records a structured post-interview scorecard containing performance ratings and feedback. The scorecard is stored against the appropriate mock interview and can be accessed by the Student Mentee.